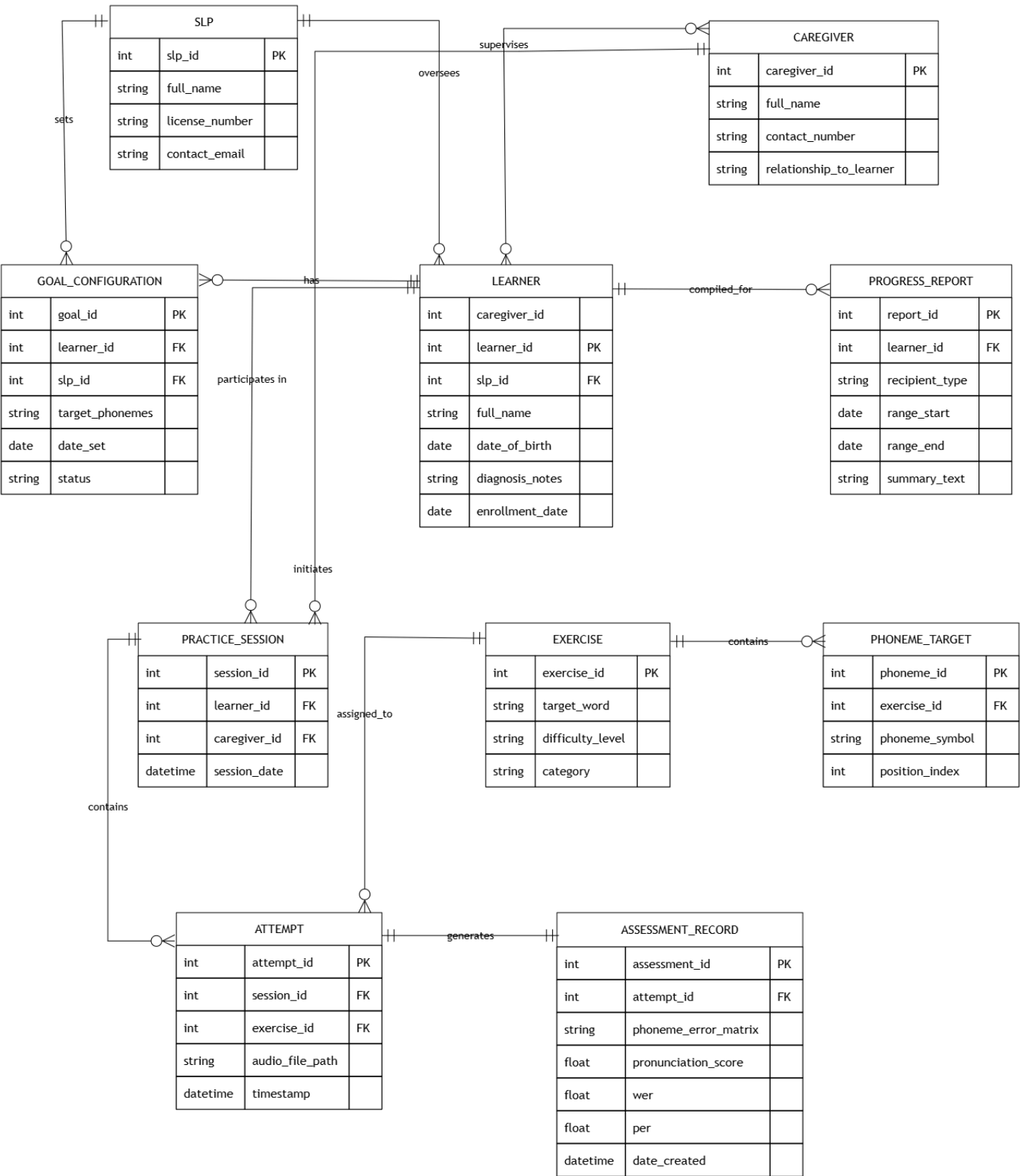


Research Title: An On-Device Speech Recognition and Pronunciation Assessment App for Listen-and-Repeat Speech Therapy

Crow’s Foot Entity-Relationship Diagram (ERD)



An Entity-Relationship Diagram (ERD) using Crow's Foot notation provides a clear visual representation of database entities and their structural relationships. Originally introduced by Dr. Gordon Everest (1976) and later popularized by James Martin through Information Engineering (IE) methodology, Crow's Foot notation is widely used in database design to model logical data structures and specify precise constraints. The notation derives its name from the three-pronged symbol used to represent the "many" side of a relationship.

Entities

In Crow's Foot ER diagrams, an entity is typically represented as a rectangle. Each rectangle contains the entity name, and sometimes the primary key is underlined.

Relationships

Relationships between entities are represented by lines connecting the rectangles. For each line:

- The end of the line often uses a set of symbols (crow's foot, line, or circle) to indicate cardinality and optional/mandatory participation.

Cardinality and Participation Symbols

Crow's Foot notation uses the following common symbols at the ends of relationship lines:

- Single line (|): Represents one entity.
- Crow's foot (three prongs): Represents many entities.
- Circle (O): Indicates zero (optional) participation.
- Combination of symbols:
 - |O: One or zero (optional one).
 - 0.. or O with crow's foot: Zero or many.
 - |< with crow's foot: One to many (mandatory many).

Example Interpretation

- A line ending with a crow's foot and a circle (O—<||—crow's foot>) signifies that an entity can be linked to zero or more instances of the related entity.
- A single line at both ends (|—|) indicates a one-to-one relationship.
- A line connecting with a single line at one end and a crow's foot at the other (|—<) indicates a one-to-many relationship.

Optionality

- Optional participation is represented with a circle at the entity end of the relationship line.
- Mandatory participation is represented with a solid line or absence of a circle.

Visual Summary

- Entity: Rectangle

- One: |
- Many: < (crow's foot)
- Optional: O
- Mandatory: Solid line interfacing entity

Entity & Field Mapping Summary

Entity	Attributes	
SLP	slp_id (PK), full_name, license_number, contact_email	Represents the licensed Speech-Language Pathologist. Sets therapy goals for learners and provides the human-in-the-loop clinical oversight the study calls for.
CAREGIVER	caregiver_id (PK), full_name, contact_number, relationship_to_learner	Represents the parent or guardian who runs practice at home. Selects exercises and initiates practice sessions on the learner's behalf.
LEARNER	learner_id (PK), slp_id (FK), full_name, date_of_birth, diagnosis_notes, enrollment_date	Represents the child receiving therapy — the central subject of the whole system. Everything (goals, sessions, progress) is tracked relative to a learner.

GOAL_CONFIGURATION	goal_id (PK), learner_id (FK), slp_id (FK), target_phonemes, date_set, status	Stores the therapy targets an SLP has configured for a specific learner (e.g. which phonemes to focus on). Corresponds to the "D2 Goal Config Store" in your DFD.
EXERCISE	exercise_id (PK), target_word, difficulty_level, category	Represents a practice item — a target word the learner is asked to repeat. Forms the content library the app draws from.
PHONEME_TARGET	phoneme_id (PK), exercise_id (FK), phoneme_symbol, position_index	Breaks an exercise word down into its individual target phonemes and their order. Corresponds to "D3 Phoneme Target Dictionary" — this is what the assessment engine aligns predicted speech against.
PRACTICE_SESSION	session_id (PK), learner_id (FK), caregiver_id (FK), exercise_id (FK), session_date	Represents one sitting of practice: a specific learner, guided by a specific caregiver, working on a specific exercise, at a specific time.
ASSESSMENT_RECORD	assessment_id (PK), attempt_id (FK), phoneme_error_matrix,	Stores the automated, on-device scoring output for one attempt

	pronunciation_score, wer, per, date_created	(phoneme-level accuracy, WER, PER). This is the core clinical data point — and what gets validated against SLP judgment per Objective 3.
PROGRESS_REPORT	report_id (PK), learner_id (FK), recipient_type, range_start, range_end, summary_text	Aggregates assessment history over a time range into a readable summary — session recaps for caregivers, longitudinal trend reports for SLPs. Implements the "Monitor Progress" process (5.0) from Level 1 DFD.